

E9 205 Machine Learning for Signal Processing

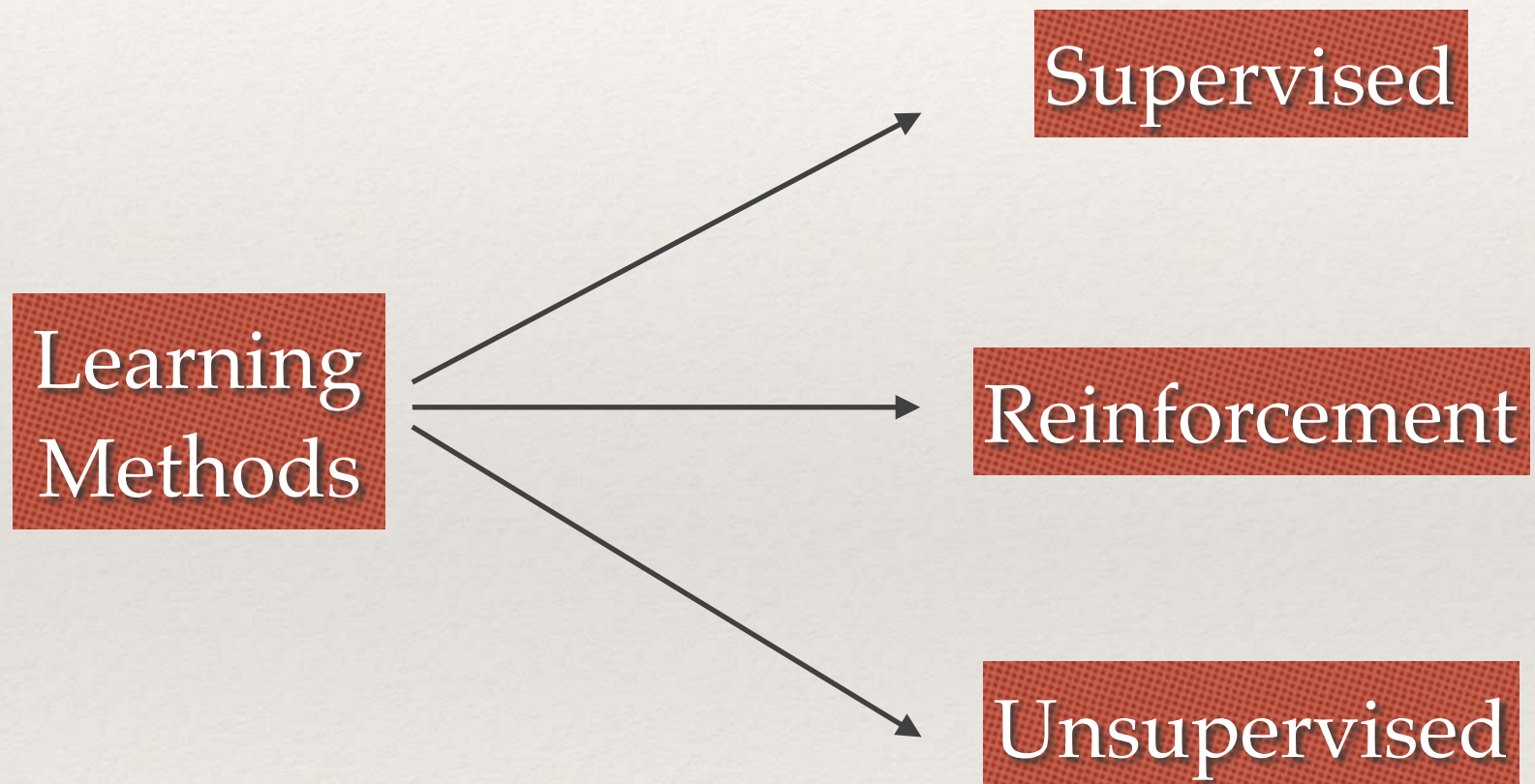
Feature Extraction

08-08-2016

Recap

- ❖ Real-world signals
 - ❖ Patterns in signal
- ❖ Learning - uncovering the underlying patterns
- ❖ Roadmap of the course

Types of Learning



Camstra, Vinciarelli, "Machine Learning for Audio, Image and Video Analysis" 2007.

Unsupervised Learning

- ❖ Data is presented without associated output targets
- ❖ Extracting structure from the data.
- ❖ Examples like clustering and segmentation.
- ❖ Concise description of the data - dimensionality reduction methods.

Reinforcement Learning

- ❖ Dynamic environment resulting in triplets - state / action / reward.
- ❖ No optimal action for a given state
- ❖ Algorithm has to learn actions in a way such the expected reward is maximized over time.
- ❖ May also involve minimizing punishment.
- ❖ Reward / punishment could be delayed - learning based on past actions.

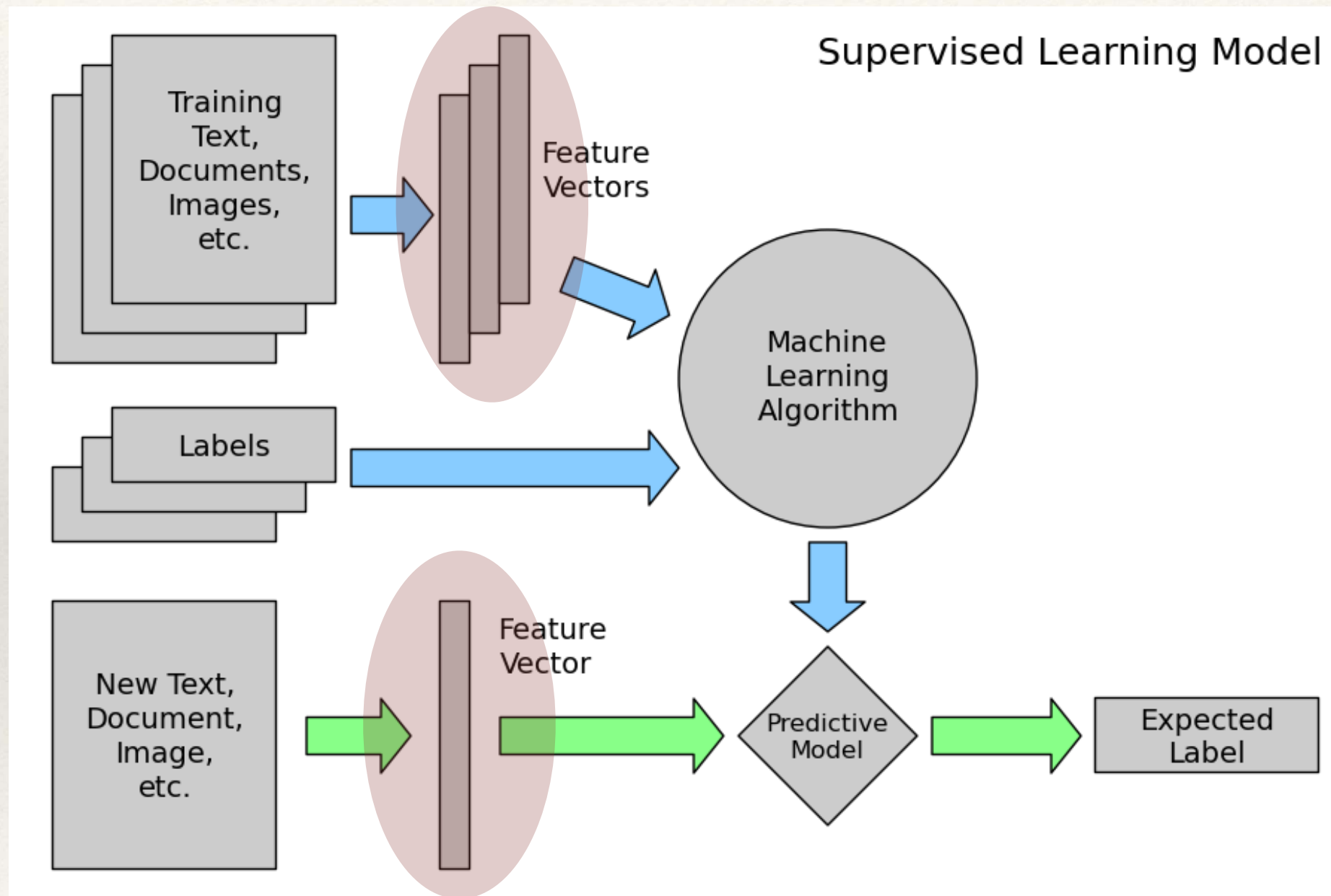
Sutton, Barto, "Reinforcement Learning: An Introduction." MIT Press, 1998.



Supervised Learning

- ❖ Training data is provided with along with target values (ground truth).
- ❖ Goal - to learn the mapping function from data to targets.
- ❖ Use the mapping function to predict unseen / test data samples.
- ❖ Two types based on the structure of the labels.
 - ❖ Classification - discrete number of classes or categories.
 - ❖ Regression - continuous output variables.

Supervised Learning



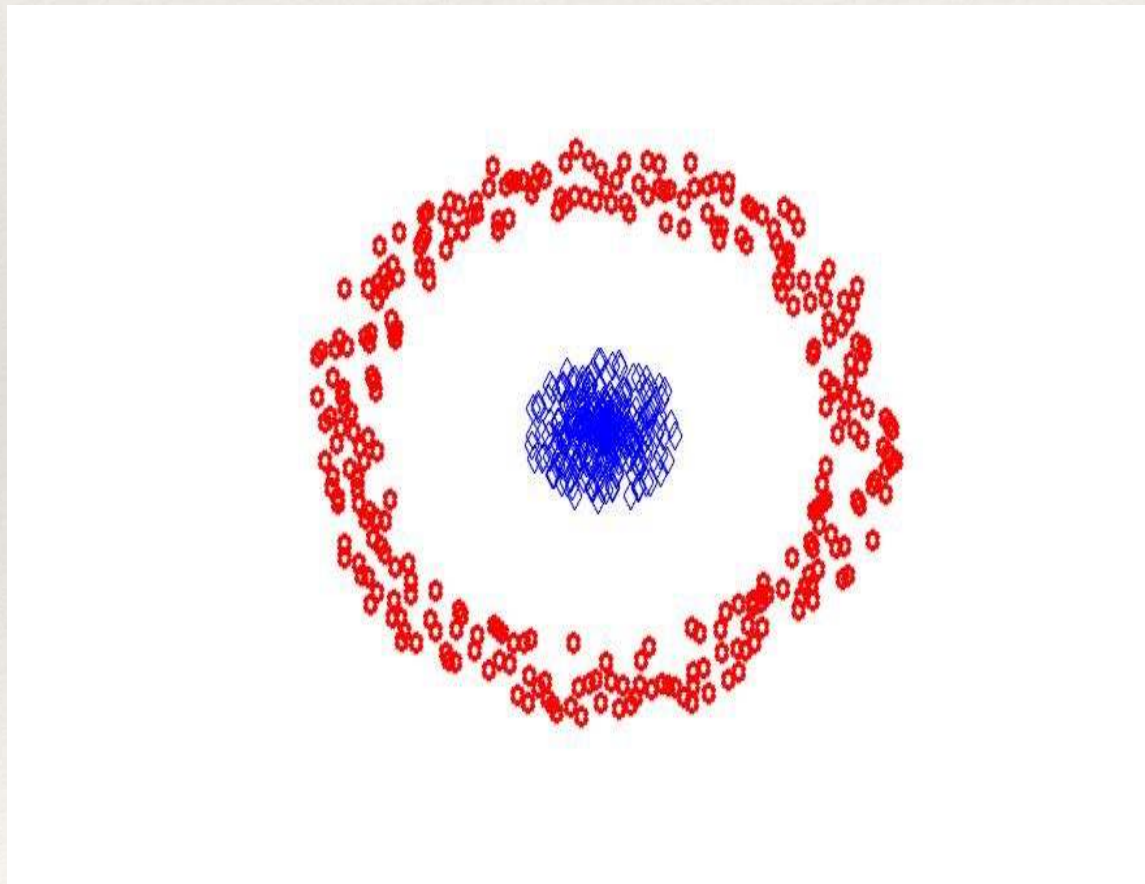
Feature Extraction

- ❖ Feature Extraction
 - ❖ Using measured data to build desirable values.
 - ❖ Attributes of the data that are informative and non-redundant.
 - ❖ Resilience to noise / artifacts.
 - ❖ Facilitating subsequent learning algorithm.

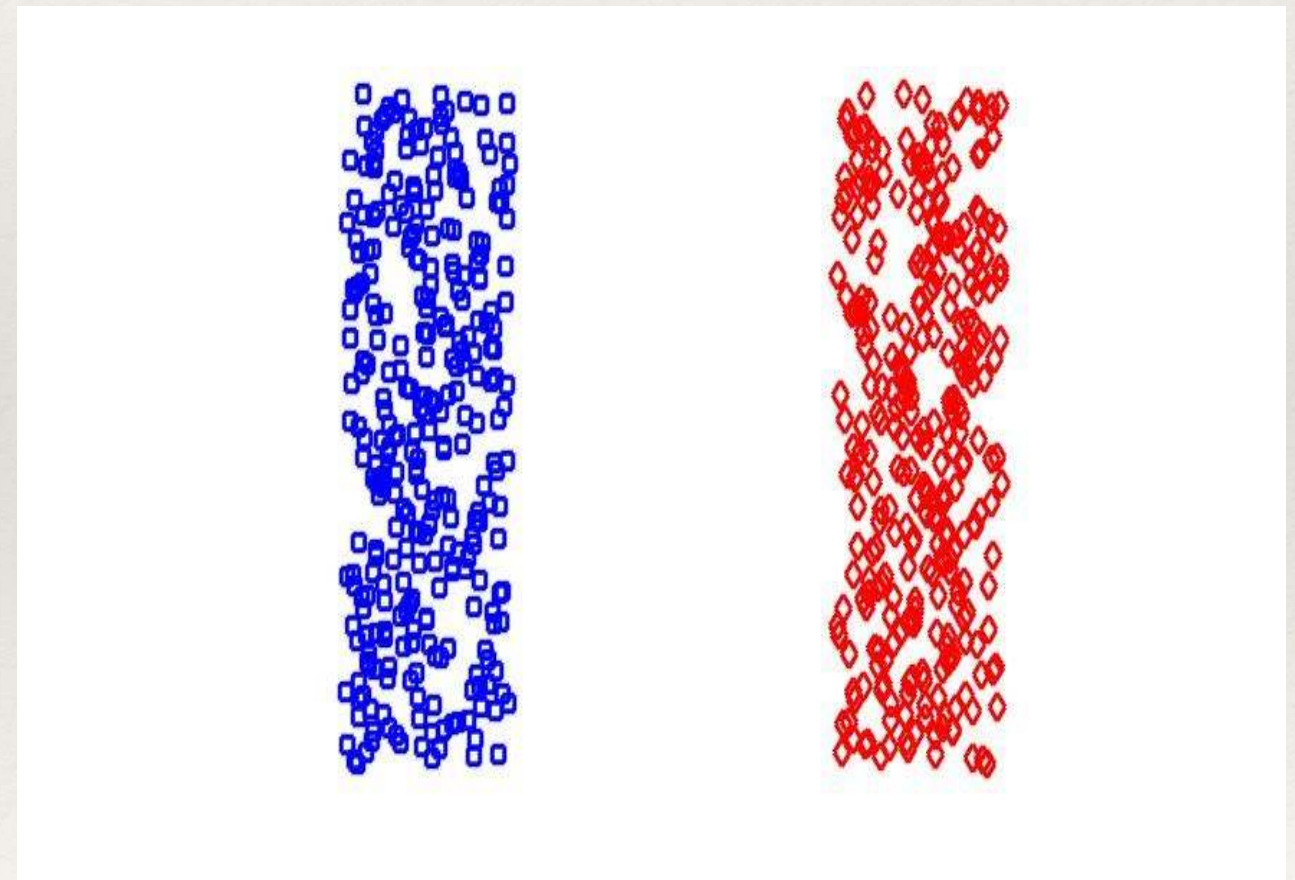
Feature Extraction

❖ Representation Problem

Cartesian Coordinates



Polar Coordinates



Feature Extraction

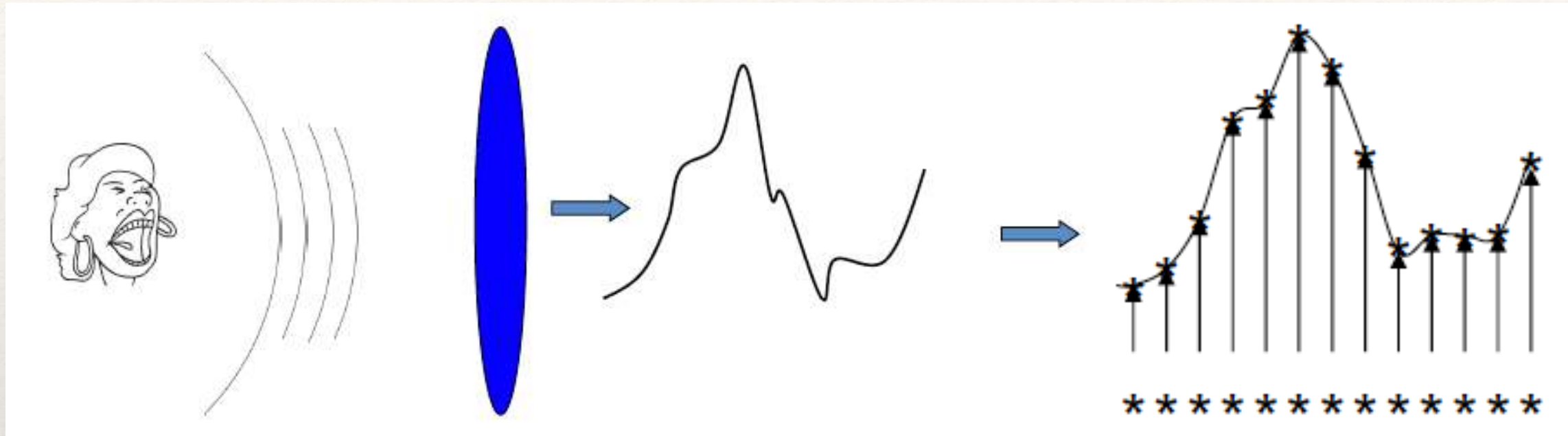
Scope for this course

- I. Feature Extraction in Speech and Audio signals.
- II. Feature Extraction Methods for Images.
- III. Brief Introduction to Feature Extraction in Text.

Speech and Audio

- ❖ Speech / Audio - 1D signals
 - ❖ Generated by pressure variations producing regions of high pressure and low pressure.
 - ❖ Travels through a medium of propagation (like air, water etc).
 - ❖ Human sensory organ - eardrum.
 - ❖ Converting pressure variations to electrical signals.
 - ❖ Action mimicked by a microphone.

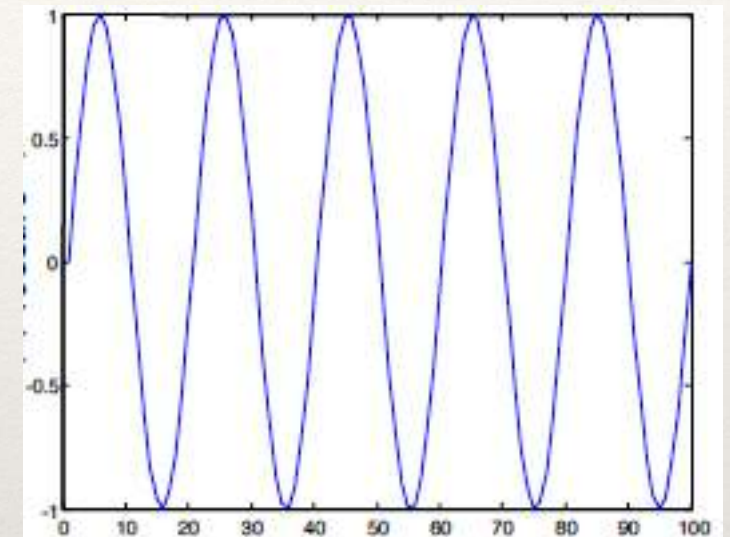
Sound waves in a computer



- ❖ Analog continuous signal from the microphone
- ❖ Discretized in time - sampling.
- ❖ Digitized in values - quantization.

Sampling

- ❖ Signals like speech / audio - analyzed in terms of sinusoids.
- ❖ Can be considered as a set of basis functions.
- ❖ Complex sinusoid - $ae^{-j2\pi f_0 t}$
- ❖ Signal expressed as weighted sum (integral) of sinusoids.
- ❖ Continuous Time Fourier Transform

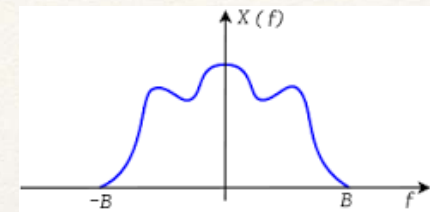


$$x(t) = \frac{1}{2\pi} \int X(f) e^{j2\pi f t} df$$

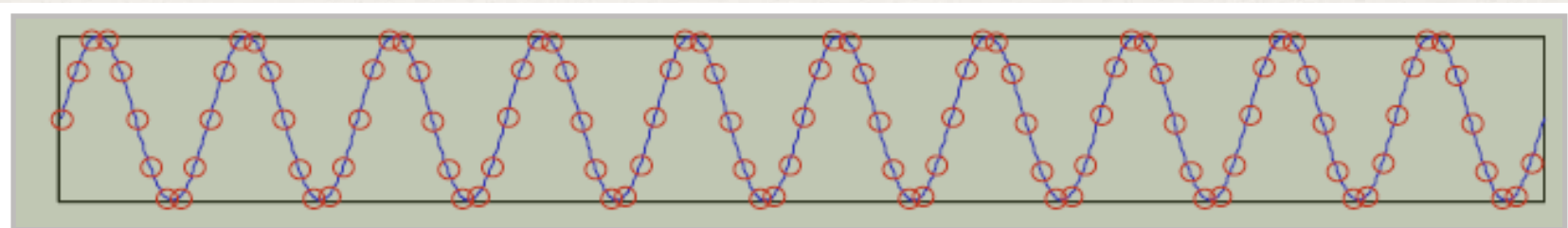
$$X(f) = \int x(t) e^{-j2\pi f t} dt$$

Sampling

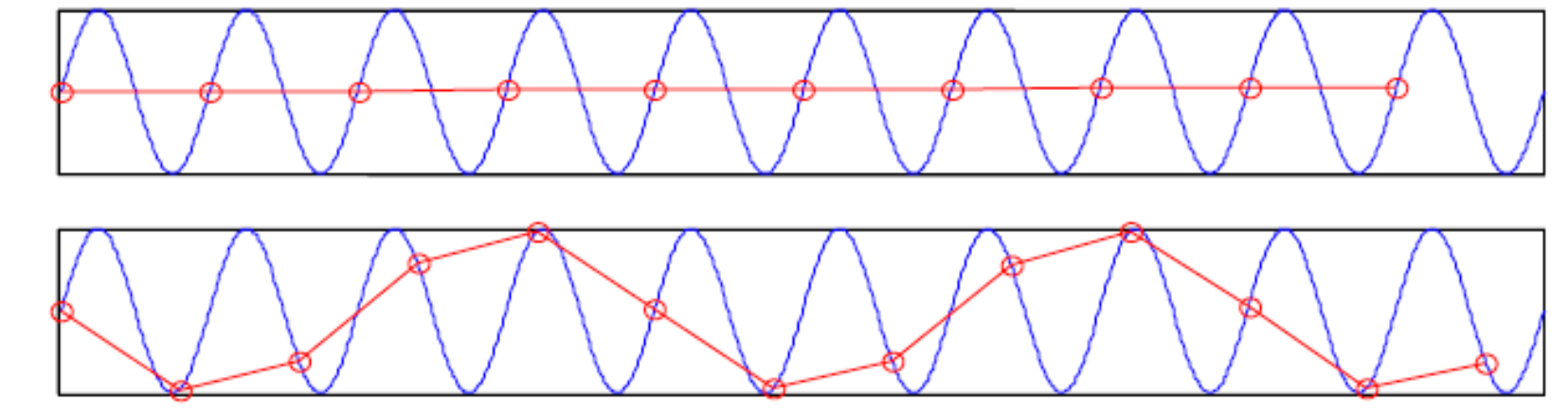
- ❖ Band limited signals $X(f) = 0 \quad |f| > B$
- ❖ Nyquist theorem - sampling frequency $f_s \geq 2B$



Oversampling



Undersampling

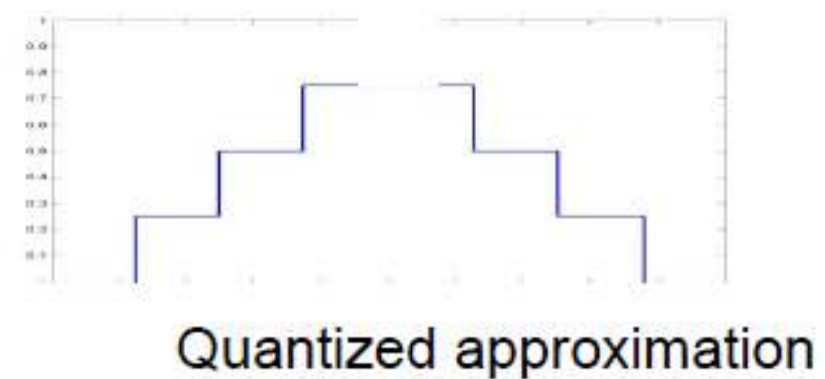
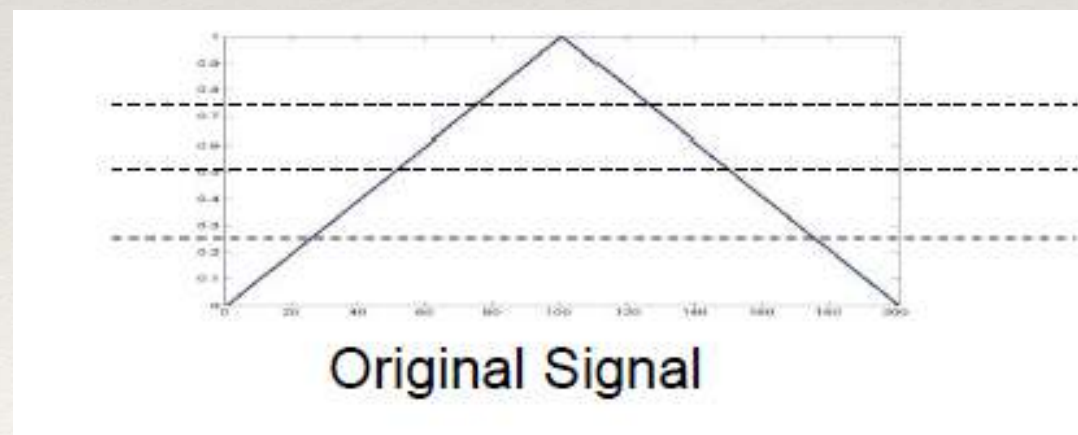
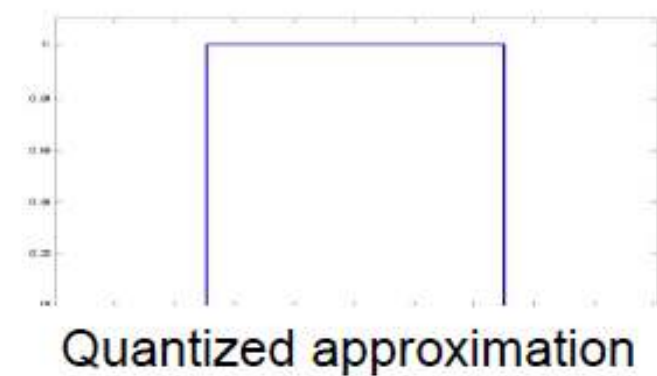
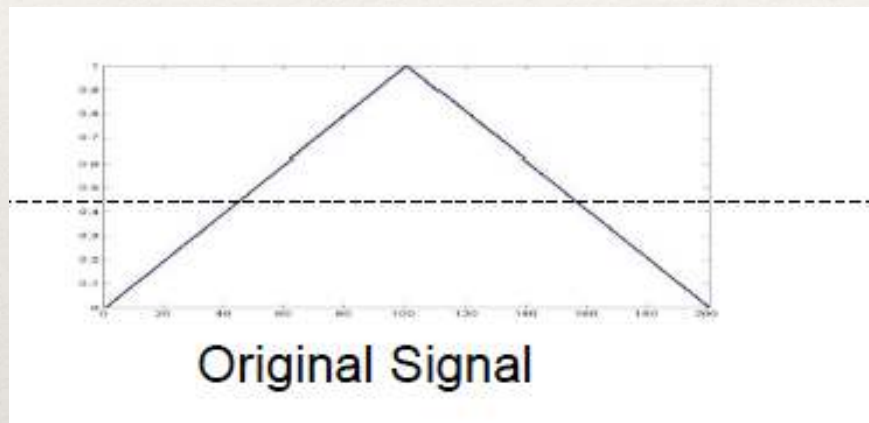


- ❖ Speech signals

- ❖ maximum frequency $\sim 4 - 8$ kHz, typical sampling frequency - (8/16 kHz).

Quantization

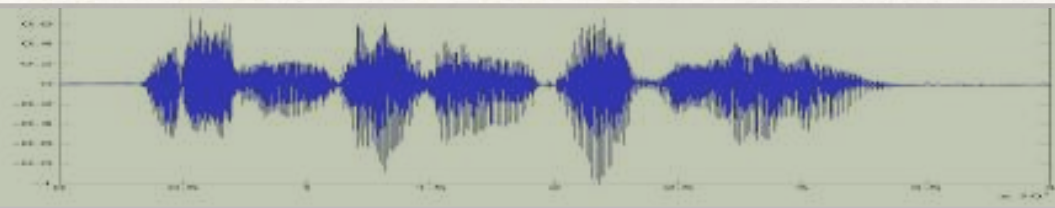
- ❖ Storing real values using finite number of bits



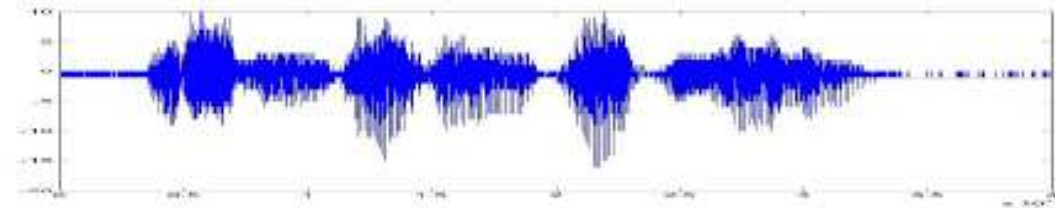
Quantization

❖ Speech signal quantization

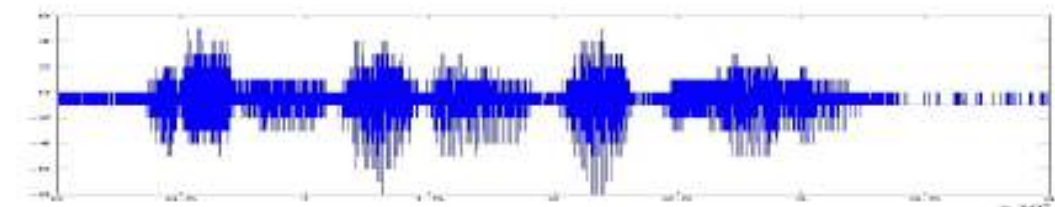
- 16 bit sampling



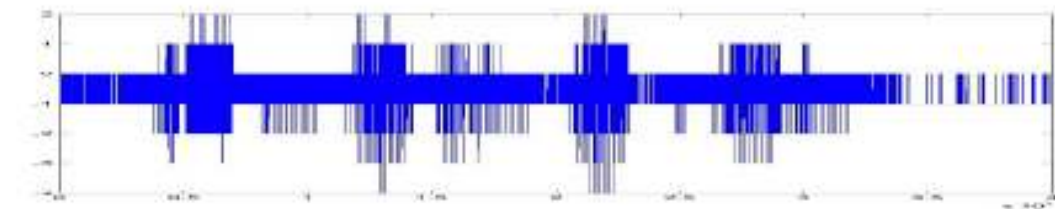
- 5 bit sampling



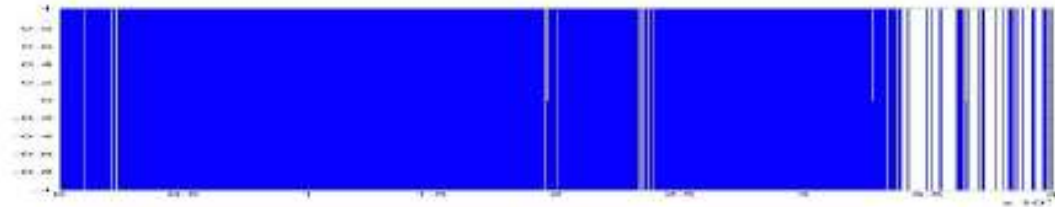
- 4 bit sampling



- 3 bit sampling

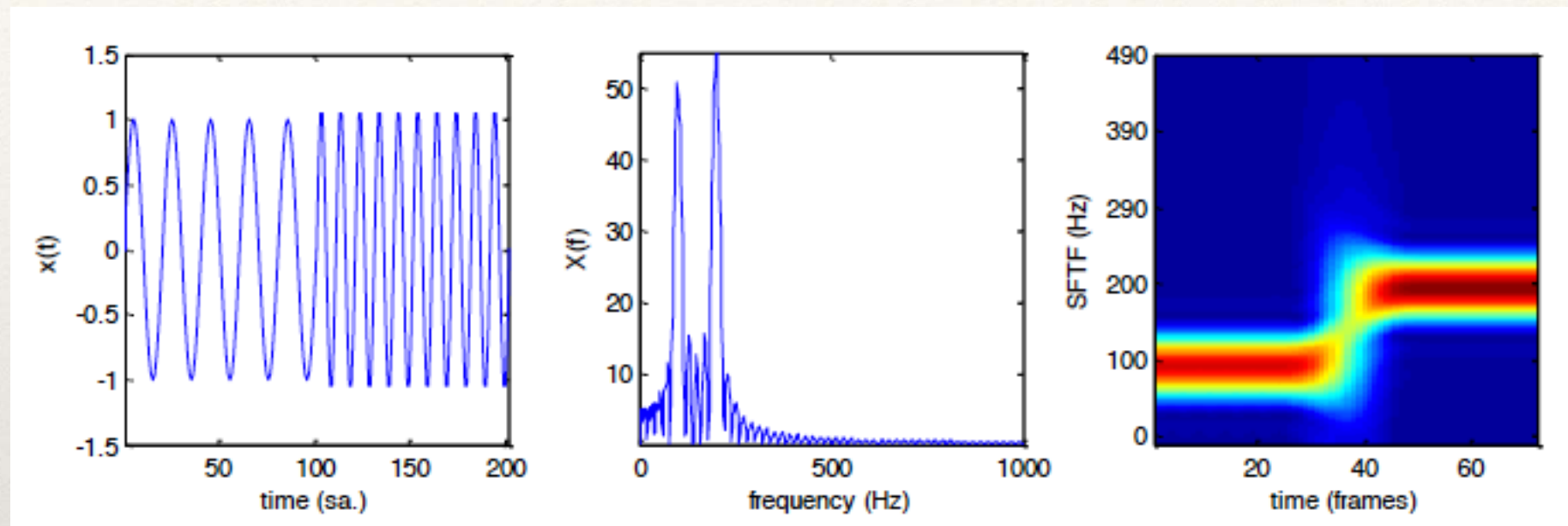


- 1 bit sampling



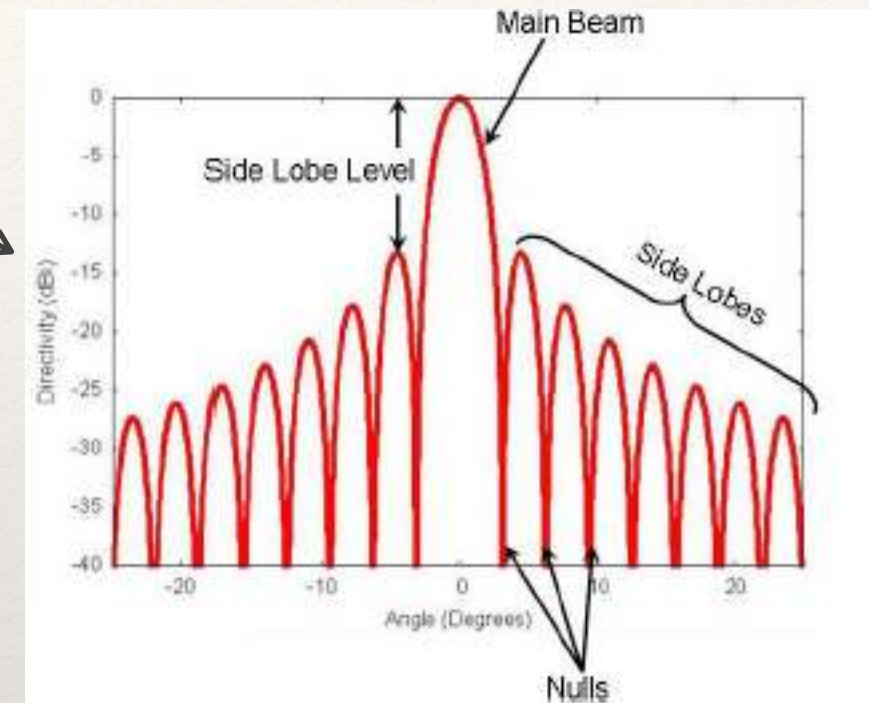
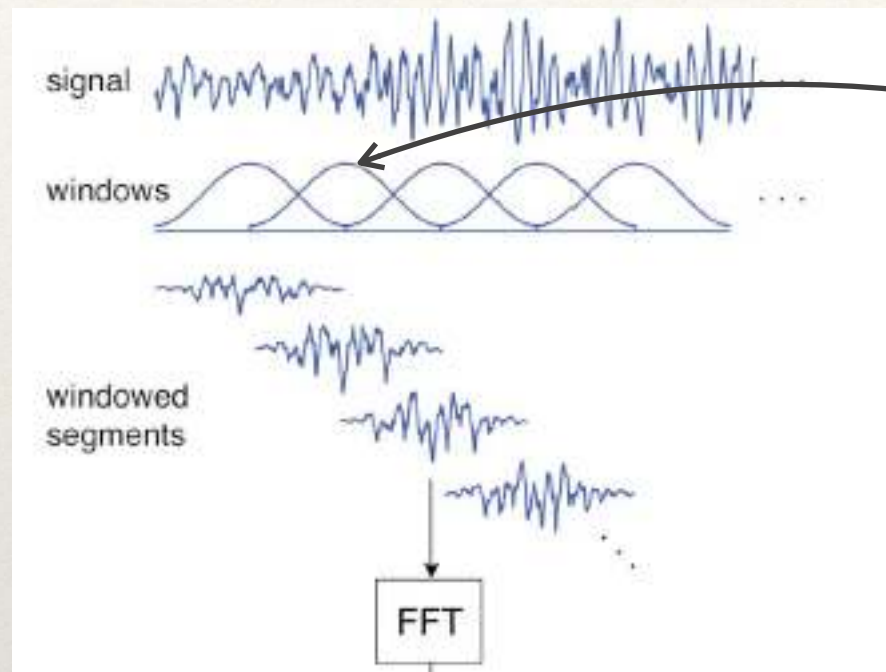
Short-term Fourier Transform

Why do we need time varying Fourier Transform



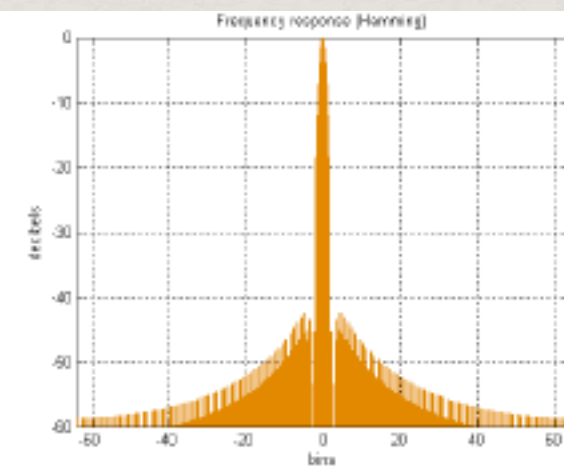
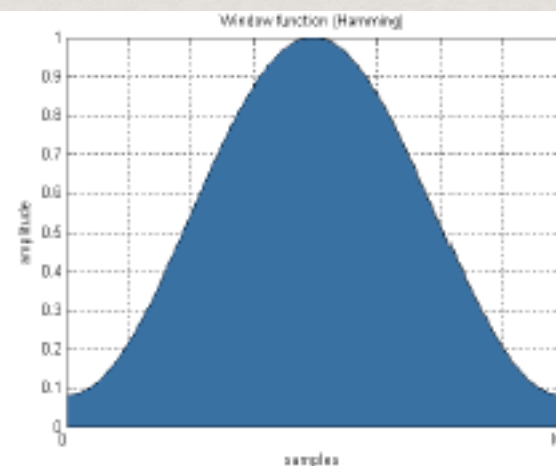
- ❖ When the signal properties change in time
- ❖ DFT will only capture the average spectral character
- ❖ Short-window analysis can indicate the change in spectrum.

Summary of STFT Properties



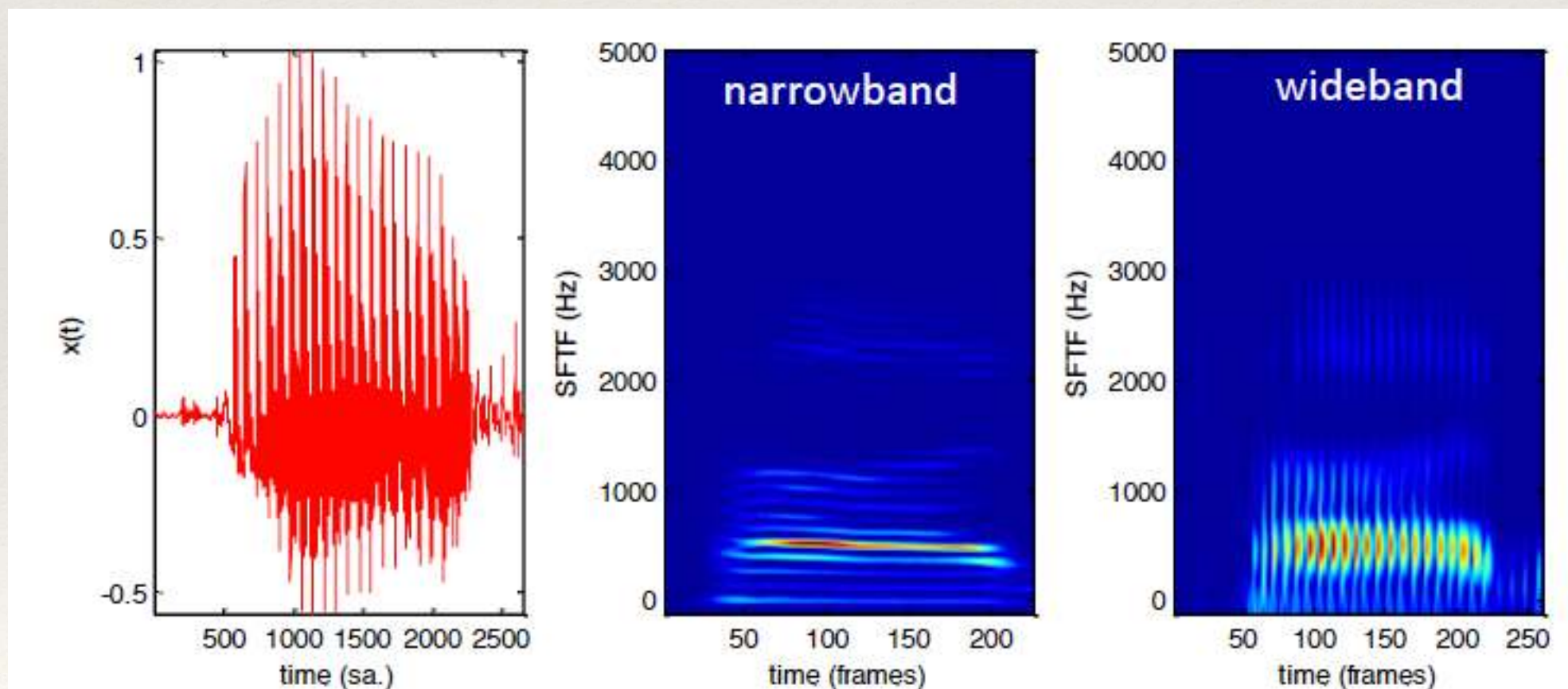
$$X[k, n_0]$$

Hamming



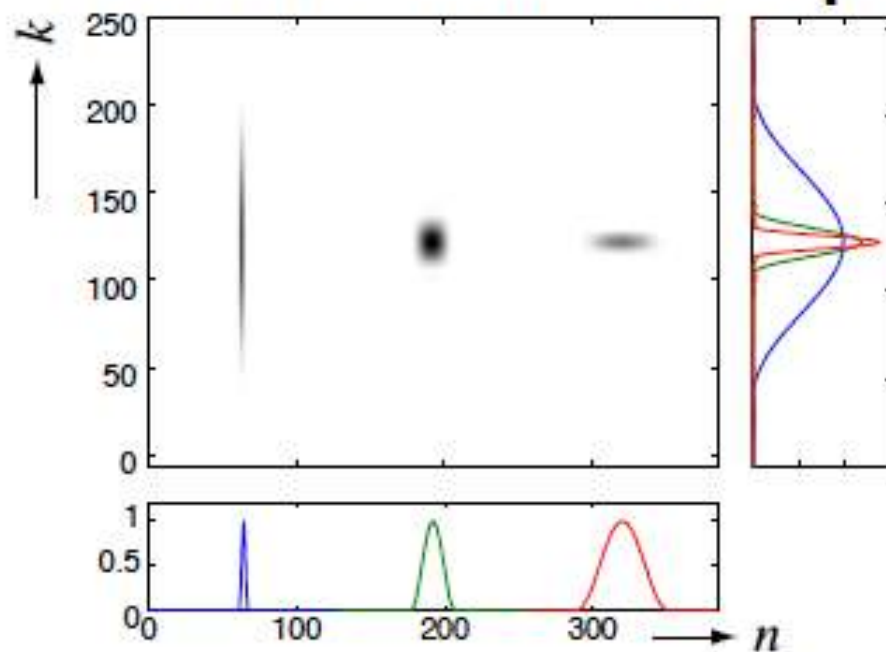
Narrowband versus Wideband

- ❖ Short windows - poor frequency resolution - wideband spectrogram
- ❖ Long windows - poor time resolution - narrowband spectrogram



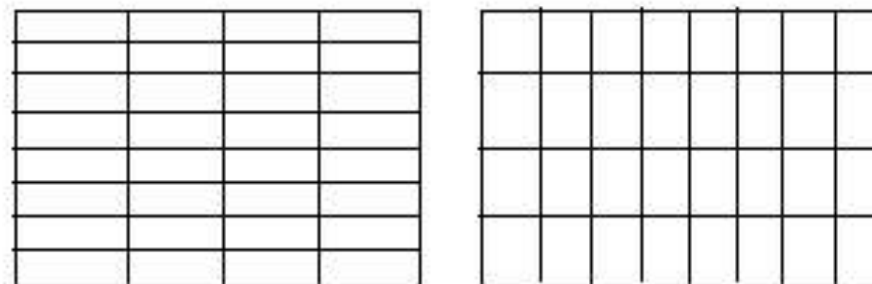
Narrowband versus Wideband

- Can illustrate time-frequency tradeoff on the time-frequency plane:

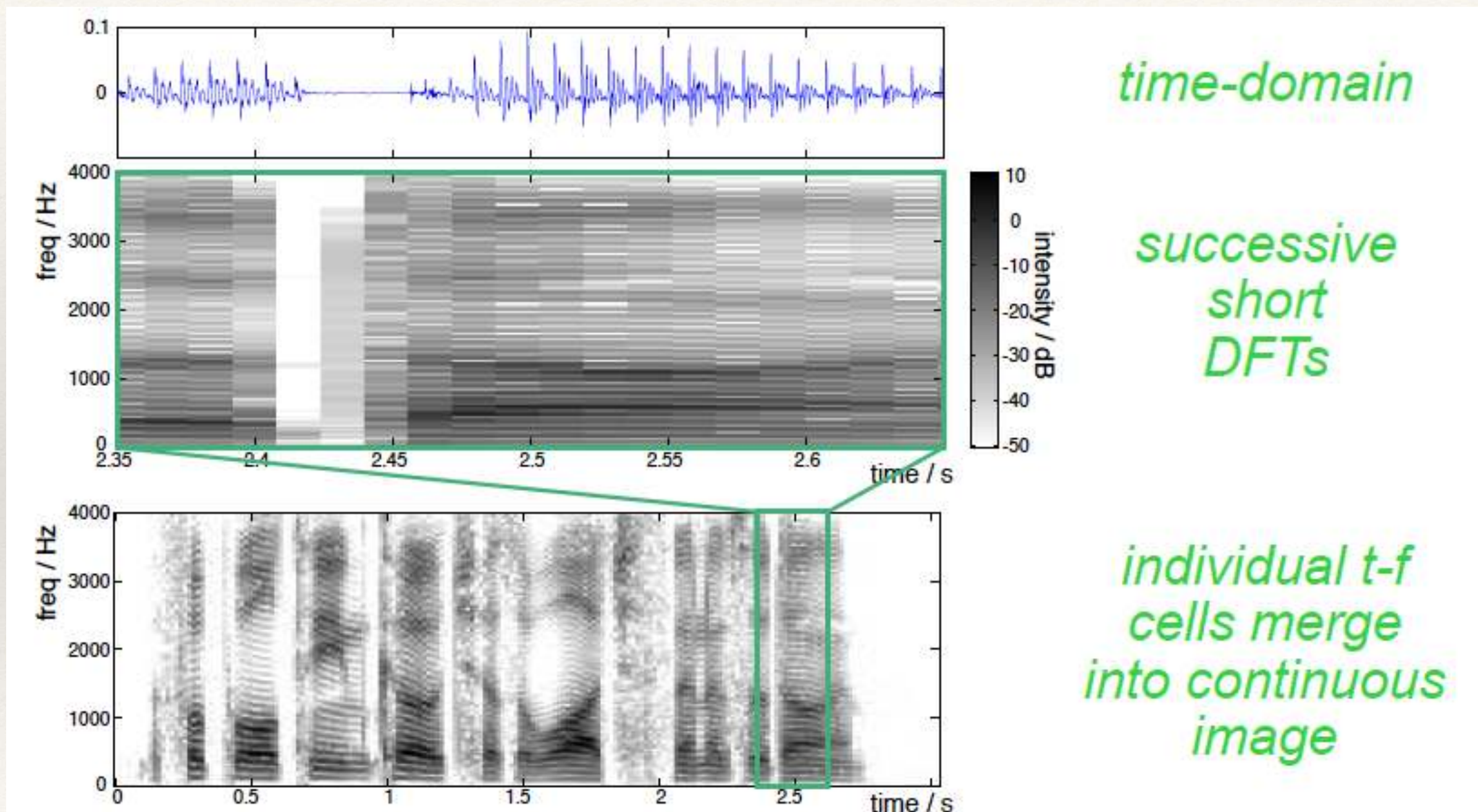


*disks show 'blurring'
due to window length;
area of disk is constant
→ Uncertainty principle:
 $\delta f \cdot \delta t \geq k$*

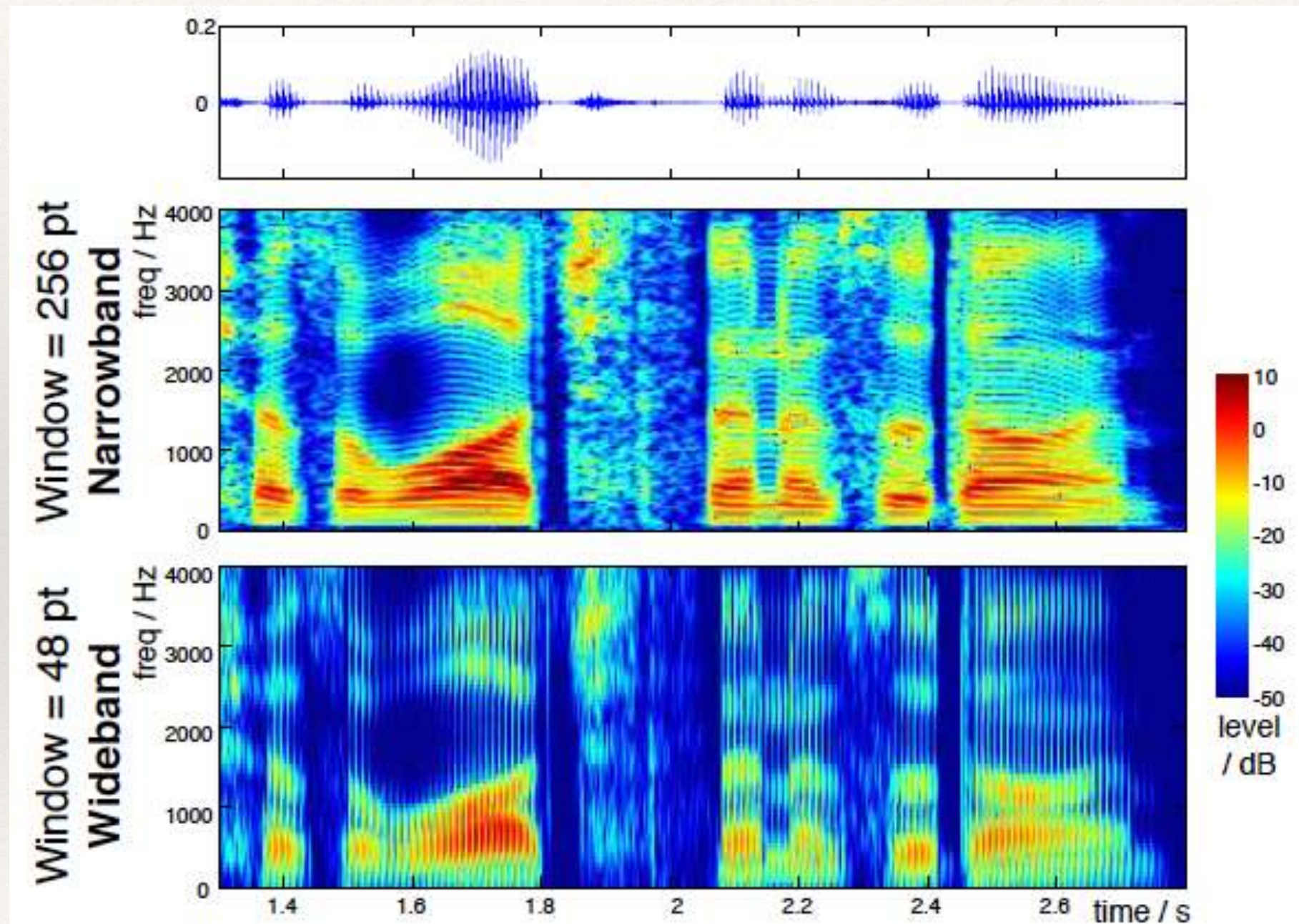
- Alternate tilings of time-freq:



Spectrogram of Real Sounds



Narrowband versus Wideband



Spectrogram in Matlab

```
>> [d,sr]=wavread('mpgr1_sx419.wav');  
>> Nw=256;  
>> specgram(d,Nw,sr)  
>> caxis([-80 0])  
>> colorbar
```

(hann) window length

*actual sampling rate
(to label time axis)*

